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Comparison of Model Selection Criteria for Neighborhood Selection Method in the Gaussian Graphical Models

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ABSTRACT

Comparison of Model Selection Criteria for Neighborhood Selection Method in the Gaussian Graphical Models

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For the gaussian graphical models(GGMs), penalized estimators are generally used in the high-dimensional situation. Representative estimator is neighborhood selection which is penalized regression based method. So there is the model selection problem for the neighborhood selection method. In this paper, we explore various model selection criteria based on the generalized information criterion(GIC) for the neighborhood selection method in the GGMs by the simulation study.

keyword : Gaussian graphical model, Model selection criterion, Neighborhood selection.

Chapter 1. Introduction

Graphical models are a probability model of multivariate variables, they are related to graph. The graphical models represent the relationships among the variables visually, it can be applied widely where the relationships are severely entangled. The relationships are expressed by edge of the graph, the edges related to the conditional dependencies between the nodes which correspond to variables.

Furthermore if the random variables follow multivariate gaussian distribution, then it is gaussian graphical models (GGMs). In the GGMs, zero entries of the inverse covariance matrix correspond to conditional dependencies between the random variables given all other variables. This problem is covariance selection (Dempster, 1972), related studies still continues these days.

Especially under the high-dimensional situation where the number of variables p is greater than the sample size n, the sample covariance matrix is singular. Therefore it is hard to get the inverse covariance matrix estimate. So various methods have been introduced for the high dimensional situation, mainly there are two approaches.

One of approaches is regression based method. Neighborhood selection method (Meinshausen and Bühlmann, 2006) proposed to identify the nonzero elements of the inverse covariance matrix by performing variable by variable LASSO regression (Tibshirani, 1996). Peng et al. (2009) proposed a joint regression model which simultaneously performs regression for all variables. Dantzig selector based estimator and CLIME estimator are suggested in Yuan (2010) and Cai et al. (2011). The proposed methods are implemented via linear programming.

Another estimation method is penalized log-likelihood method which directly estimates the inverse covariance matrix. Yuan and Lin (2007) proposed penalized likelihood method for model selection and estimation. To solve the optimization problem, they used maxdet algorithm. Banerjee et al. (2008) solved the optimization problem of the estimator which was proposed Yuan

and Lin (2007) by using block coordinate descent algorithm. Based on approach in Banerjee et al. (2008), Friedman, Hastie, and Tibshirani (2008) proposed the Graphical LASSO (GLASSO) algorithm. Mazumder et al. (2012) analyze the theoretical properties of GLASSO. Under the penalized likelihood, non-convex penalties also have been considered by Fan and Feng (2009) instead of L1-penalty.

Most of the estimators mentioned above need to select the tuning parameters. Depending on which tuning parameter is selected, the complexity of the graph is different. Thus, tuning parameter selection is also important issue in the GGMs. However, there are few studies on the tuning parameters compared to estimation. Foygel and Drton (2010) studied the consistency of extended BIC (EBIC) for GGMs where both p and sample size increase. Their simulation gives the results of GLASSO with EBIC. Gao et al. (2012) studied the consistency of the BIC for penalized MLE with SCAD. And they also considered the consistency of modified BIC when p tends to infinity as sample size increases. Liu and Wang (2017) proposed TIGER which used penalized regression with SQRT-LASSO. It is asymptotically tuning free and non-asymptotically tuning insensitive. Recently, an estimator based on the generalized information criterion (Konishi and Kitagawa, 1996) has been proposed in Abbruzzo et al. (2019).

In this paper, we focus on the comparing the model selection criteria based on the generalized information criterion (Konishi and Kitagawa, 1996; Kim et al., 2012) for the neighborhood selection method in the GGMs. For this purpose, first, we review the graphical models and the gaussian graphical models which follow the multivariate gaussian distribution. Next, we introduce the neighborhood selection method which is the penalized regression based estimator. Also we introduce the model selection criterion for the neighborhood selection. In addition, we provide simulation studies to compare the various information criteria for the neighborhood selection.

The remain of this paper is organized as follows. Chapter 2, we briefly introduce the graphical models. Also we review the gaussian graphical models which is the representative graphical models. Chapter 3, we describe regression based estimator and its model selection criteria. Chapter 4, we give some

simulation studies comparing with the performance of the neighborhood selection method with different model selection criteria.

Chapter 2. Graphical Models: Review

In this section, we review the graphical models briefly. For the full details about the theory, recommend to see the following papers: Lauritzen (1996), Drton and Maathuis (2017), and Maathuis et al. (2018).

1. Graphical Models

A graphical model is useful tool which can represent the conditional dependencies among the random variables by using the graph (Lauritzen, 1996). Let $G = (V, E)$ be a graph, where $V = \{1, \ldots, p\}$ is the set of nodes and E is the set of edges in $V \times V$. In the graphical models, the nodes correspond to random variables and the edges represent the conditional dependencies between the variables (Drton and Maathuis, 2017). The graphical models can be either directed or undirected, depending on which way the edge is pointing. In other words, suppose the adjacency matrix $A = \{a_{ij}\}$, if the A is symmetric(i.e., $a_{ij} = a_{ji}$), it is an undirected graph and if $a_{ij} \neq a_{ji}$, it is directed graph. Therefore if we look at the relationship between variables through an undirected graph, it is called an undirected graphical models. While if we represent the relationship associated with causality through the directed graph, it is referred to a directed graphical models.

When an undirected graph is given, then we can estimate the structure of undirected graph. Briefly there are two types of graph which is the marginal correlation graphs and the conditional independent graphs. The marginal correlation graph is the correlation between two variables. On the other hand, the conditional independent graph is the partial correlation between two variables given a set of random variables. In this paper, we focus on the undirected graphical models and the conditional independent graph.

Let $X = (X_1, \ldots, X_p)^T$ be a p-dimensional random vector correspond to the set of nodes V in the undirected graph G. A random vector X satisfies three different Markov properties with respect to G (Lauritzen, 1996; Bühlmann and

Van De Geer, 2011; Drton and Maathuis, 2017): pairwise, local, and global.

A random vector X satisfies the pairwise Markov property relative to G if

$$X_i \perp X_j | X_{V \setminus \{i,j\}}$$

for all $(i,j) \notin E, i \neq j$, where $V \setminus \{i,j\}$ denote that belong to set of nodes V which do not contain nodes i and j. And we can say that X satisfies the local Markov property relative to G if

$$X_i \perp \{X_k; k \in V \setminus \{i, a \in N(i)\}\} | X_{N(i)} \tag{1}$$

for all $k \in V$, where $N(i)$ is the set of neighbors of node i. And all nodes included in $N(i)$ are connected to node i. Let A, B and C be the subsets of V, X satisfies global Markov property relative to G if for any disjoint subsets (A, B, C) such that C separates A from B in G

$$X_A \perp X_B | X_C.$$

The three different Markov properties are equivalent and the proposition is proved in Lauritzen (1996).

2. Gaussian Graphical Models

One of the graphical models is Gaussian graphical models (GGMs). Suppose X be a p-dimensional random vector with $X \sim N(0, \Sigma)$, where Σ is the symmetric positive definite matrix, and $\Sigma^{-1} = \Theta = (\theta_{ij})$ is the inverse covariance matrix. The GGMs are a conditional independent graph with a multivariate gaussian distribution. In the GGMs, it is well known that the inverse covariance matrix corresponds to conditional independence between the random variables. The density function of X is

$$\begin{aligned} f(X; \mu = 0, \Theta) &= (2\pi)^{-p/2} \det(\Theta)^{1/2} \exp\left(-\frac{1}{2} X^T \Theta X\right) \\ &= (2\pi)^{-p/2} \det(\Theta)^{1/2} \exp\left(-\frac{1}{2} \sum_i \theta_{ii} x_i^2 - \sum_{i<j} \theta_{ij} x_i x_j\right) \\ &\propto \prod_i \exp\left(\theta_{ii} x_i^2\right) \prod_{i<j} \exp\left(\theta_{ij} x_i x_j\right). \end{aligned}$$

A multivariate gaussian distribution factorizes with respect to an undirected graph G. Under a positive distribution, factorization relative to G implies the pairwise Markov property with respect to G by Hammersley-Clifford theorem

Theorem 1 *(Hammersley and Clifford) Suppose $X \in \mathbb{R}^p$ is a random vector with positive density function with respect to a product measure. X satisfies the pairwise Markov property with respect to an undirected graph G if only and if X factorizes according to the graph G.*

From the multivariate gaussian distribution, terms of single variable X_i are the node potential. The pairs term of X_i and X_j is edge potential which corresponds to the off-diagonal of inverse covariance matrix, θ_{ij}. GGMs are represented by the node and edge potentials. In addition, if $\theta_{ij} = 0$, then there is no edge.

Let the covariance matrix and its inverse be a block matrix and partition X into X_i and a $(p-1)$ vector X_j,

$$\Sigma = \begin{pmatrix} \sigma_{ii} & \Sigma_{ij} \\ \Sigma_{ij}^T & \Sigma_{jj} \end{pmatrix}, \quad \Theta = \begin{pmatrix} \theta_{ii} & \Theta_{ij} \\ \Theta_{ij}^T & \Theta_{jj} \end{pmatrix}.$$

By the Lemma of block matrix inversion,

$$\theta_{ii} = (\sigma_{ii} - \Sigma_{ij}\Sigma_{jj}^{-1}\Sigma_{ij}^T)^{-1}. \tag{2}$$

Next, a multivariate gaussian distribution has a property as follows. The conditional distribution of X_i given the rest follows a normal with density is

$$X_i | X_j \sim N(\Sigma_{ij}\Sigma_{jj}^{-1}X_j,\ \Sigma_{ii} - \Sigma_{ij}\Sigma_{jj}^{-1}\Sigma_{ji}).$$

The conditional distribution also corresponds to linear regression coefficients. Suppose the true linear regression model for a random vector X is given as

$$X_i = \sum_{j \in V \setminus \{i\}} \beta_{ij} X_j + \epsilon_i,$$

where X_i is a i-th column vector of X and it is dependent variable, $X_j = \{X_k; k \in V \setminus \{i\}\}$ is a independent variable, $\beta_j \in \mathbb{R}^{p-1}$ is a regression coefficient, and ϵ_i is independent and identically distributed $N(0, \sigma_i^2)$. The regression function also can be written as

$$\beta_{ij} = \Sigma_{ij}\Sigma_{jj}^{-1}, \quad Var(\epsilon_i) = \sigma_{ii} - \Sigma_{ij}\Sigma_{jj}^{-1}\Sigma_{ij}^T. \tag{3}$$

By the block matrix inversion (2) and the property of conditional distribution (3), we can take

$$\begin{aligned}
\theta_{ii} &= (\sigma_{ii} - \Sigma_{ij}\Sigma_{jj}^{-1}\Sigma_{ij}^{T})^{-1} \\
&= \{Var(\epsilon_i)\}^{-1}, \\
\Theta_{ij} &= -(\sigma_{ii} - \Sigma_{ij}\Sigma_{jj}^{-1}\Sigma_{ij}^{T})^{-1}\Sigma_{ij}\Sigma_{jj}^{-1} \\
&= -\{Var(\epsilon_i)\}^{-1}\Sigma_{ij}\Sigma_{jj}^{-1} \\
&= -\theta_{ii}\beta_{ij}.
\end{aligned}$$

Therefore

$$\beta_{ij} = -\Theta_{ij}\theta_{ii}^{-1}. \tag{4}$$

Hence (4) implies that β_{ij} depends on the nonzero off-diagonal element of the inverse covariance matrix. And $\beta_{ij} = 0$ implies $X_i \perp X_j | X_{V\setminus\{i,j\}}$. That is the zero element of the inverse covariance matrix corresponds to the conditionally independent (Uhler, 2017),

$$(i,j) \notin E \iff X_i \perp X_j | X_{V\setminus\{i,j\}} \iff \Theta_{ij} = \Theta_{ji} = 0 \iff \beta_{ij} = \beta_{ji} = 0. \tag{5}$$

In summary, non-edges in an undirected graph G, the conditional independent, zero elements in the inverse covariance matrix, and zero regression coefficients are equivalent in the GGMs through (5) (Uhler, 2017).

Chapter 3. Estimation of the conditional independence

Representative estimator with GGMs is neighborhood selection method (Meinshausen and Bühlmann, 2006) which is regression based estimator. Neighborhood selection performs penalized regression separately column by column. By performing the penalized regression, we can estimate conditional independence. Also through the existence of the edge, it is possible to identify the graphical model. In this section, we introduce the neighborhood selection models briefly.

1. Neighborhood selection

Neighborhood selection method estimates the zeros in the inverse covariance matrix by fitting a penalized regression that each variable used as the response and the rest used as the predictors. The goal of neighborhood selection is estimating the neighborhoods of given variables. Neighborhood selection considers the Local Markov property and the multivariate gaussian distribution property.

First, we recall $N(i)$ which is the set of neighbors of a node i. All nodes included in $N(i)$ are connected to node i. When a node i's neighborhood $N(i)$ is given, node i is conditional independent all the other variables except node i and $N(i)$, i.e. (1). From (3), the nonzero regression coefficients for the true regression model of X_i is same for the variables that contained in the set of neighborhoods, so the set of neighborhoods $N(i)$ can be written as

$$N(i) = \{j \in V : \beta_{ij} \neq 0\}. \tag{6}$$

(6) implies that $N(i)$ includes the best predictor of X_i with response variable X_i and predictors $\{X_j : j \in V \backslash \{i\}\}$ in linear regression (Meinshausen and Bühlmann, 2006; Maathuis et al., 2018).

Therefore neighborhood selection method estimates the set of neighborhoods $N(i)$. When the observations are given, the penalized regression estimate $\hat{\beta}_{ij}$ is given by

$$\hat{\beta}_{ij} = \arg\min_{\beta_{ij}} \left(\frac{1}{n} ||X_i - \sum_{j \neq i} X_j \beta_{ij}||_2^2 + \sum_{j \neq i} J(|\beta_{ij}|) \right),$$

where J is a penalty function with a tuning parameter.

And the nonzero $\hat{\beta}_{ij}$ corresponds to the nonzero off-diagonal element of the precision matrix. The neighborhood estimate is written as

$$\hat{N}(i) = \{j \in V : \hat{\beta}_{ij} \neq 0\}.$$

The rule for estimating the $\hat{E}$ follows AND/OR (Meinshausen and Bühlmann, 2006),

- OR rule:
$$\hat{E}^{or} = \{(i,j) : i \in \hat{N}(i) \vee j \in \hat{N}(i)\}$$
- AND rule:
$$\hat{E}^{and} = \{(i,j) : i \in \hat{N}(i) \wedge j \in \hat{N}(i)\},$$

where $i \neq j$.

Originally, Meinshausen and Bühlmann (2006) used the LASSO which is the convex problem and makes the regression coefficient exactly zero. Other penalty has been considered in Wang et al. (2016). They proposed non-convex penalty based method. In this paper, we consider the LASSO and SCAD which is the non-convex problem.

2. Model selection criterion for neighborhood selection

As model selection is required for high dimensional data modeling, various model selection criteria have been proposed. The well-known model selection criteria such as the AIC (Akaike, 1998), the BIC (Schwarz, 1978) are introduced for a long time. Cross-validation also commonly used for the simplicity but it is known to overfit as the AIC.

The generalized information criterion (GIC) has been considered in Konishi and Kitagawa (1996), and Kim et al. (2012). The GIC family includes not only AIC, BIC but also other information criteria. Especially, Kim et al. (2012) studied sufficient conditions of many information criteria based on the GIC.

The tuning parameter controls the sparsity of the estimator in the neighborhood selection. To choose the tuning parameter, we used the model selection criterion based on the GIC. When σ is unknown, we may use the mean squared error of the full model. In this case, the GIC for the neighborhood selection is defined as

$$\mathrm{GIC}_{\lambda}^{NS}(\mathcal{S}) = \log(||X_i - \mathbf{X}_{\mathcal{S}}\hat{\beta}_{\mathcal{S}}||^2/n) + \frac{\lambda}{n}|\mathcal{S}|, \tag{7}$$

where $\mathcal{S}$ is a subset of $\{1, \ldots, i-1, i+1, \ldots, p\}$, $\mathbf{X}_{\mathcal{S}}$ is a sub-matrix of X_{-i} that consists of columns $X_k, k \in \mathcal{S}$, $\beta_{\mathcal{S}}$ is a sub-vector of β that consists of $\beta_k, k \in \mathcal{S}$, $\hat{\beta}_{\mathcal{S}} = \arg\min_{\beta_{\mathcal{S}}} ||X_i - \mathbf{X}_{\mathcal{S}}\beta_{\mathcal{S}}||_2^2 + J(\beta_{\mathcal{S}})$, $||\cdot||$ is the Euclidean norm, and λ is a preset positive sequence of the sample size n and model size p. The weight varies depending on the λ and n. The final estimator from the GIC is defined as

$$\hat{\mathcal{S}} = \underset{\mathcal{S} \in \{1,\ldots,i-1,i+1,\ldots,p\}}{\mathrm{argmin}} \mathrm{GIC}_{\lambda}^{NS}(\mathcal{S}).$$

Chapter 4. Numerical Study

In this section, we present the results of simulation studies according to the model selection criteria AIC$(= \text{GIC}_1)$, BIC$(= \text{GIC}_2)$, GIC_3, GIC_4, GIC_5, GIC_6, and GIC_7 for the neighborhood selection with the LASSO and SCAD. We obtained all the penalized estimators using R package glmnet and ncvreg.

1. Simulation studies

The simulation studies was based on the multivariate gaussian distribution with two graphical models. In Model 1, we made the matrix structure at random. Model 2 has been considered in Yuan and Lin (2007).

- Model 1. Let $K = (k_{ij})$ be a p by p matrix, where $i < j$. We choose a nonzero positions of k_{ij} by $\{(p-1)p/2\} * 0.05$ randomly. The signals of nonzero elements are given in sequences whose minimum value is 0.3 and maximum value is 0.6. And then make K as symmetric matrix. Next, we calculate the absolute of minimum eigenvalue $\min(\mathbf{\Lambda})$ of K. Then we set the diagonal elements of inverse covariance matrix as $\Theta^0_{ii} = (|\min(\mathbf{\Lambda})| + 0.05)\mathbf{I}_p$ for the symmetric positive definite (Cai et al., 2016) and the off-diagonal elements as $\Theta^0_{ij} = K$.
- Model 2. AR(2): $\Theta^0_{i,i} = 1, \Theta^0_{i,i-1} = \Theta^0_{i-1,i} = 0.5$, and $\Theta^0_{i,i-2} = \Theta^0_{i-2,i} = 0.25$

Each model, we generated n random samples of X from $N(0, (\Theta^0)^{-1})$. And we set $n \in \{100, 150, 200, 250, 300, 350, 400\}, p \in \{100, 150, 200, 250, 300, 350, 400\}$ for each simulation. We repeated the simulation 200 times.

The main goal of the simulation is to compare the model selection criteria based on the GIC for the neighborhood selection. We compare the finite sample performance of the GICs which have been considered in Kim et al. (2012), λs associated with GIC are given as

- AIC$(= \text{GIC}_1) : \lambda_1 = 2$,

- BIC$(= \text{GIC}_2) : \lambda_2 = \log(n)$,
- $\text{GIC}_3 : \lambda_3 = p^{1/3}$,
- $\text{GIC}_4 : \lambda_4 = 2\log(p)$,
- $\text{GIC}_5 : \lambda_5 = 2\{\log(p) + \log\log(p)\}$,
- $\text{GIC}_6 : \lambda_6 = \log\{\log(n)\}\log(p)$,
- $\text{GIC}_7 : \lambda_7 = \log(n)\log(p)$.

We calculated two measures for checking the performance: sensitivity and specificity which are defined as:

$$\text{Sensitivity (SEN)} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{Specificity (SPC)} = \frac{\text{TN}}{\text{TN} + \text{FP}},$$

where TP = $\sum_{i<j} I(\hat{\Theta}_{ij} \neq 0, \Theta_{ij} \neq 0)$, FP = $\sum_{i<j} I(\hat{\Theta}_{ij} \neq 0, \Theta_{ij} = 0)$, TN = $\sum_{i<j} I(\hat{\Theta}_{ij} = 0, \Theta_{ij} = 0)$, and FN = $\sum_{i<j} I(\hat{\Theta}_{ij} = 0, \Theta_{ij} \neq 0)$.

Tables and figures summarize the results. We first discuss SEN of Model 1:

- For the GIC_3,GIC_4,GIC_5,GIC_6, and GIC_7, SEN increases as n increases in both the LASSO and SCAD. However, as p increases while maintaining n fixed, SEN decreases. As GIC_3 has lower weight, it has higher SEN than GIC_4,GIC_5,GIC_6, and GIC_7. On the other hand, GIC_7 which has the largest weight, has the lowest SEN. Among the LASSO and SCAD, the SCAD produces the larger SEN than the LASSO. One characteristic is that GIC_3 has the larger SEN when $n = p$, particularly at $n < 200$.

- For the AIC, the SEN increases as n increases in both the LASSO and SCAD. In the case of LASSO, SEN decreases starting with $n = p$. In the case of SCAD, SEN decreases up to $n > p$, increases where $n = p$, and decreases again when $n < p$. Among the LASSO and SCAD, the LASSO produces the larger SEN than SCAD.

- For the BIC, in the case of LASSO, SEN increases where $n = p$ when $n \leq 200$. The SEN increases as n increases when $n \geq 250$. In the case of SCAD, on the other hand, SEN increases as n increases overall. But $n \leq 300$, SEN decreases up to $n > p$, increases where $n = p$. Among the

LASSO and SCAD, the SCAD produces the larger SEN than the LASSO except $n \leq 150$.

Second, we discuss SPC of Model 1:

- For the GIC_5 and GIC_7, there is no change with SPC converging to 1. For the GIC_4 and GIC_6, also SPC converging to 1 when the weight which depends on n and p is larger. This implies SPC converges to 1 if the weight is greater than a certain strength. Strength is different on the situation, because it depends on n and p.
- For the AIC, while SEN converges to 1, SPC is lower than other GICs. This implies AIC tends to overfit. And if $n > p$, based on the same situation as n and p, SPC decreases. But if $n < p$, SPC increases.
- For the BIC and GIC_3, overall, SPC increases as n increases in both the LASSO and SCAD. However, SPC drops when $n = p$ abruptly. Among the LASSO and SCAD, the LASSO produces the larger SPC than the SCAD except for the case of $n \leq 150$ in BIC.

Next, the results in the AR(2) model are not very different patterns from the Model 1. One difference is SEN converge to 1 faster than Model 1. For the details, refer to <Table 4-5>, <Table 4-6>, <Table 4-7>, <Table 4-8>, <Figure 4-5>, <Figure 4-6>, <Figure 4-7>, and <Figure 4-8>.

To sum up, (a) for all GICs, SEN increases as increases in both the LASSO and SCAD. Also, the larger weight of the GIC, the lower SEN and the SEN increases slowly even if n increases. And in GIC_3, GIC_4, GIC_5, GIC_6, and GIC_7, SCAD has the higher SEN than LASSO. (b) SPC converges to 1 if the weight is greater than a certain strength which depends on n and p. However, for the AIC, BIC, and GIC_3, SPC decreases when $n = p$.

<Table 4-1>: Average of the sensitivity with the LASSO under Model 1 situation

		Sensitivity						
n	p	AIC	BIC	GIC_3	GIC_4	GIC_5	GIC_6	GIC_7
100	100	1.000	0.958	0.956	0.385	0.265	0.510	0.089
	150	0.888	0.783	0.472	0.163	0.092	0.254	0.020
	200	0.821	0.802	0.319	0.078	0.039	0.138	0.007
	250	0.763	0.757	0.169	0.040	0.020	0.076	0.003
	300	0.708	0.705	0.089	0.022	0.010	0.045	0.001
	350	0.659	0.657	0.053	0.013	0.006	0.028	0.001
	400	0.615	0.614	0.034	0.008	0.004	0.019	0.000
150	100	0.972	0.799	0.818	0.610	0.484	0.692	0.186
	150	1.000	0.877	0.805	0.347	0.222	0.446	0.047
	200	0.913	0.507	0.429	0.178	0.101	0.259	0.018
	250	0.868	0.561	0.290	0.102	0.053	0.159	0.007
	300	0.825	0.684	0.186	0.058	0.029	0.097	0.004
	350	0.784	0.720	0.123	0.037	0.018	0.063	0.002
	400	0.747	0.715	0.082	0.024	0.011	0.042	0.001
200	100	0.980	0.876	0.896	0.754	0.654	0.802	0.296
	150	0.981	0.760	0.759	0.514	0.368	0.598	0.090
	200	1.000	0.768	0.635	0.301	0.183	0.392	0.031
	250	0.926	0.511	0.432	0.180	0.100	0.254	0.015
	300	0.890	0.420	0.310	0.114	0.059	0.169	0.007
	350	0.857	0.364	0.212	0.071	0.036	0.109	0.004
	400	0.826	0.395	0.149	0.048	0.023	0.075	0.002
250	100	0.989	0.919	0.936	0.849	0.778	0.875	0.417
	150	0.976	0.831	0.840	0.650	0.516	0.712	0.138
	200	0.986	0.727	0.710	0.442	0.293	0.523	0.051
	250	1.000	0.680	0.564	0.276	0.162	0.356	0.023
	300	0.932	0.526	0.433	0.185	0.101	0.251	0.013
	350	0.904	0.434	0.316	0.116	0.061	0.166	0.007
	400	0.877	0.363	0.230	0.080	0.041	0.117	0.004
300	100	0.994	0.951	0.964	0.906	0.856	0.921	0.535
	150	0.983	0.881	0.891	0.751	0.640	0.793	0.197
	200	0.975	0.797	0.792	0.566	0.411	0.635	0.076
	250	0.992	0.701	0.666	0.391	0.244	0.469	0.035
	300	1.000	0.633	0.540	0.260	0.146	0.333	0.017
	350	0.937	0.528	0.422	0.179	0.097	0.237	0.011
	400	0.913	0.454	0.325	0.125	0.065	0.171	0.007
350	100	0.996	0.966	0.976	0.935	0.903	0.946	0.637
	150	0.988	0.916	0.926	0.824	0.736	0.852	0.265
	200	0.978	0.845	0.846	0.667	0.520	0.719	0.104
	250	0.979	0.771	0.751	0.506	0.343	0.575	0.049
	300	0.995	0.688	0.636	0.356	0.215	0.429	0.025
	350	1.000	0.609	0.518	0.243	0.133	0.308	0.015
	400	0.939	0.526	0.410	0.172	0.091	0.224	0.009
400	100	0.998	0.978	0.985	0.956	0.933	0.963	0.707
	150	0.992	0.937	0.947	0.869	0.805	0.888	0.337
	200	0.983	0.882	0.886	0.746	0.621	0.783	0.138
	250	0.977	0.817	0.806	0.596	0.432	0.652	0.066
	300	0.983	0.741	0.705	0.441	0.276	0.508	0.034
	350	0.997	0.674	0.606	0.325	0.187	0.391	0.019
	400	1.000	0.604	0.503	0.234	0.125	0.292	0.012

<Table 4-2>: Average of the sensitivity with the SCAD under Model 1 situation

		Sensitivity						
n	p	AIC	BIC	GIC_3	GIC_4	GIC_5	GIC_6	GIC_7
100	100	0.991	0.975	0.974	0.451	0.309	0.724	0.138
	150	0.741	0.725	0.705	0.203	0.126	0.319	0.040
	200	0.630	0.620	0.605	0.105	0.058	0.252	0.016
	250	0.540	0.531	0.518	0.055	0.029	0.225	0.008
	300	0.469	0.460	0.447	0.032	0.015	0.205	0.003
	350	0.413	0.404	0.392	0.018	0.009	0.185	0.002
	400	0.367	0.358	0.345	0.012	0.005	0.163	0.001
150	100	0.924	0.800	0.819	0.635	0.525	0.702	0.247
	150	0.991	0.954	0.929	0.396	0.278	0.525	0.083
	200	0.772	0.737	0.629	0.228	0.139	0.314	0.032
	250	0.693	0.678	0.603	0.136	0.076	0.205	0.015
	300	0.626	0.615	0.553	0.079	0.041	0.128	0.007
	350	0.563	0.554	0.496	0.049	0.025	0.085	0.004
	400	0.509	0.500	0.442	0.032	0.015	0.058	0.002
200	100	0.959	0.871	0.891	0.768	0.683	0.808	0.368
	150	0.912	0.772	0.772	0.562	0.435	0.635	0.136
	200	0.991	0.924	0.839	0.362	0.239	0.449	0.052
	250	0.788	0.705	0.510	0.232	0.139	0.311	0.025
	300	0.728	0.699	0.443	0.152	0.084	0.216	0.013
	350	0.671	0.654	0.395	0.095	0.050	0.141	0.007
	400	0.618	0.606	0.353	0.063	0.031	0.098	0.004
250	100	0.978	0.918	0.933	0.859	0.803	0.881	0.499
	150	0.940	0.836	0.843	0.692	0.580	0.741	0.206
	200	0.909	0.749	0.733	0.507	0.370	0.579	0.083
	250	0.990	0.883	0.728	0.343	0.220	0.423	0.038
	300	0.797	0.648	0.501	0.241	0.140	0.314	0.021
	350	0.747	0.686	0.387	0.154	0.084	0.214	0.011
	400	0.700	0.671	0.299	0.106	0.055	0.152	0.007
300	100	0.989	0.949	0.960	0.913	0.875	0.926	0.619
	150	0.962	0.884	0.893	0.786	0.699	0.815	0.282
	200	0.926	0.810	0.805	0.630	0.498	0.684	0.120
	250	0.911	0.731	0.701	0.468	0.322	0.536	0.057
	300	0.990	0.835	0.640	0.331	0.202	0.405	0.027
	350	0.803	0.611	0.491	0.235	0.135	0.302	0.016
	400	0.760	0.637	0.394	0.166	0.089	0.222	0.010
350	100	0.994	0.967	0.974	0.943	0.918	0.950	0.710
	150	0.976	0.918	0.926	0.850	0.785	0.871	0.363
	200	0.947	0.855	0.856	0.723	0.608	0.760	0.160
	250	0.918	0.792	0.776	0.582	0.436	0.637	0.077
	300	0.915	0.724	0.682	0.439	0.290	0.506	0.040
	350	0.989	0.783	0.591	0.314	0.185	0.381	0.022
	400	0.804	0.598	0.482	0.228	0.126	0.288	0.013
400	100	0.997	0.978	0.984	0.963	0.945	0.966	0.784
	150	0.984	0.939	0.946	0.891	0.844	0.904	0.450
	200	0.963	0.890	0.893	0.790	0.700	0.817	0.213
	250	0.937	0.834	0.825	0.667	0.533	0.709	0.102
	300	0.912	0.773	0.744	0.528	0.366	0.585	0.051
	350	0.920	0.716	0.661	0.409	0.257	0.474	0.029
	400	0.988	0.744	0.573	0.304	0.174	0.367	0.017

<Table 4-3>: Average of the specificity with the LASSO under Model 1 situation

		Specificity						
n	p	AIC	BIC	GIC_3	GIC_4	GIC_5	GIC_6	GIC_7
100	100	0.000	0.121	0.126	0.994	0.998	0.974	1.000
	150	0.244	0.500	0.908	0.998	0.999	0.995	1.000
	200	0.371	0.414	0.927	0.999	1.000	0.998	1.000
	250	0.466	0.477	0.976	1.000	1.000	0.999	1.000
	300	0.537	0.542	0.995	1.000	1.000	0.999	1.000
	350	0.592	0.594	0.998	1.000	1.000	0.999	1.000
	400	0.635	0.637	0.999	1.000	1.000	1.000	1.000
150	100	0.375	0.965	0.959	0.990	0.995	0.984	0.999
	150	0.000	0.354	0.522	0.995	0.998	0.992	1.000
	200	0.205	0.968	0.987	0.998	0.999	0.996	1.000
	250	0.298	0.788	0.992	0.999	1.000	0.997	1.000
	300	0.376	0.593	0.995	0.999	1.000	0.999	1.000
	350	0.441	0.536	0.997	1.000	1.000	0.999	1.000
	400	0.494	0.539	0.998	1.000	1.000	0.999	1.000
200	100	0.625	0.964	0.954	0.986	0.992	0.981	0.999
	150	0.178	0.970	0.970	0.992	0.996	0.988	1.000
	200	0.000	0.619	0.875	0.996	0.998	0.993	1.000
	250	0.187	0.982	0.988	0.998	0.999	0.996	1.000
	300	0.259	0.983	0.992	0.999	1.000	0.997	1.000
	350	0.323	0.958	0.995	0.999	1.000	0.998	1.000
	400	0.380	0.881	0.997	0.999	1.000	0.999	1.000
250	100	0.693	0.962	0.950	0.983	0.989	0.978	0.998
	150	0.534	0.967	0.964	0.989	0.994	0.985	0.999
	200	0.066	0.972	0.975	0.994	0.997	0.991	1.000
	250	0.000	0.822	0.976	0.996	0.999	0.994	1.000
	300	0.182	0.982	0.989	0.998	0.999	0.996	1.000
	350	0.235	0.986	0.993	0.999	0.999	0.998	1.000
	400	0.290	0.988	0.995	0.999	1.000	0.998	1.000
300	100	0.721	0.961	0.947	0.980	0.987	0.976	0.998
	150	0.651	0.965	0.961	0.987	0.992	0.983	0.999
	200	0.402	0.970	0.971	0.991	0.996	0.988	1.000
	250	0.027	0.975	0.979	0.995	0.998	0.992	1.000
	300	0.000	0.931	0.985	0.997	0.999	0.995	1.000
	350	0.178	0.983	0.990	0.998	0.999	0.997	1.000
	400	0.223	0.985	0.993	0.998	0.999	0.998	1.000
350	100	0.737	0.962	0.947	0.979	0.986	0.975	0.997
	150	0.702	0.964	0.958	0.984	0.991	0.981	0.999
	200	0.597	0.968	0.967	0.989	0.994	0.986	1.000
	250	0.265	0.972	0.975	0.993	0.997	0.990	1.000
	300	0.013	0.976	0.982	0.995	0.998	0.993	1.000
	350	0.000	0.970	0.987	0.997	0.999	0.995	1.000
	400	0.175	0.983	0.991	0.998	0.999	0.997	1.000
400	100	0.742	0.962	0.945	0.978	0.985	0.974	0.997
	150	0.720	0.963	0.956	0.982	0.989	0.979	0.999
	200	0.671	0.966	0.965	0.987	0.993	0.984	0.999
	250	0.521	0.970	0.972	0.991	0.995	0.988	1.000
	300	0.152	0.974	0.979	0.994	0.997	0.992	1.000
	350	0.007	0.977	0.984	0.996	0.998	0.994	1.000
	400	0.000	0.979	0.988	0.997	0.999	0.996	1.000

<Table 4-4>: Average of the specificity with the SCAD under Model 1 situation

		Specificity						
n	p	AIC	BIC	GIC_3	GIC_4	GIC_5	GIC_6	GIC_7
100	100	0.008	0.055	0.057	0.931	0.996	0.566	0.999
	150	0.496	0.526	0.561	0.995	0.998	0.974	1.000
	200	0.601	0.617	0.639	0.998	0.999	0.946	1.000
	250	0.671	0.685	0.701	0.999	1.000	0.925	1.000
	300	0.721	0.733	0.748	0.999	1.000	0.919	1.000
	350	0.756	0.768	0.782	1.000	1.000	0.920	1.000
	400	0.785	0.796	0.809	1.000	1.000	0.926	1.000
150	100	0.404	0.941	0.922	0.987	0.994	0.978	0.999
	150	0.009	0.116	0.182	0.992	0.997	0.913	1.000
	200	0.486	0.570	0.762	0.996	0.998	0.991	1.000
	250	0.563	0.592	0.707	0.997	0.999	0.994	1.000
	300	0.620	0.639	0.717	0.999	1.000	0.996	1.000
	350	0.667	0.682	0.745	0.999	1.000	0.997	1.000
	400	0.703	0.716	0.773	0.999	1.000	0.998	1.000
200	100	0.501	0.956	0.939	0.985	0.992	0.978	0.999
	150	0.334	0.948	0.948	0.989	0.995	0.983	1.000
	200	0.010	0.201	0.409	0.994	0.997	0.986	1.000
	250	0.488	0.698	0.953	0.996	0.999	0.993	1.000
	300	0.547	0.608	0.920	0.998	0.999	0.995	1.000
	350	0.594	0.623	0.892	0.999	0.999	0.997	1.000
	400	0.634	0.654	0.885	0.999	1.000	0.998	1.000
250	100	0.551	0.962	0.946	0.984	0.991	0.979	0.999
	150	0.444	0.957	0.953	0.987	0.994	0.982	1.000
	200	0.282	0.954	0.961	0.991	0.996	0.987	1.000
	250	0.011	0.317	0.670	0.995	0.998	0.991	1.000
	300	0.502	0.852	0.977	0.996	0.999	0.994	1.000
	350	0.540	0.673	0.982	0.998	0.999	0.996	1.000
	400	0.581	0.636	0.983	0.998	0.999	0.997	1.000
300	100	0.590	0.967	0.953	0.984	0.990	0.980	0.999
	150	0.493	0.961	0.955	0.985	0.992	0.981	0.999
	200	0.402	0.959	0.961	0.989	0.995	0.984	1.000
	250	0.239	0.958	0.968	0.992	0.997	0.989	1.000
	300	0.012	0.458	0.867	0.995	0.998	0.992	1.000
	350	0.510	0.931	0.982	0.996	0.999	0.994	1.000
	400	0.544	0.798	0.987	0.998	0.999	0.996	1.000
350	100	0.618	0.971	0.958	0.985	0.990	0.982	0.999
	150	0.528	0.963	0.956	0.984	0.991	0.980	0.999
	200	0.456	0.961	0.961	0.987	0.993	0.983	1.000
	250	0.369	0.960	0.966	0.990	0.995	0.987	1.000
	300	0.203	0.961	0.972	0.993	0.997	0.990	1.000
	350	0.013	0.601	0.951	0.995	0.998	0.993	1.000
	400	0.517	0.956	0.984	0.997	0.999	0.995	1.000
400	100	0.639	0.974	0.962	0.985	0.990	0.983	0.999
	150	0.552	0.965	0.958	0.984	0.990	0.981	0.999
	200	0.489	0.962	0.960	0.985	0.992	0.982	1.000
	250	0.429	0.961	0.965	0.988	0.994	0.985	1.000
	300	0.337	0.963	0.970	0.992	0.996	0.989	1.000
	350	0.175	0.964	0.976	0.994	0.997	0.991	1.000
	400	0.014	0.721	0.976	0.996	0.998	0.994	1.000

<Table 4-5>: Average of the sensitivity with the LASSO under AR(2) situation

		Sensitivity						
n	p	AIC	BIC	GIC_3	GIC_4	GIC_5	GIC_6	GIC_7
100	100	1.000	0.975	0.974	0.465	0.316	0.589	0.058
	150	0.964	0.884	0.668	0.393	0.255	0.508	0.033
	200	0.952	0.948	0.623	0.338	0.207	0.452	0.021
	250	0.942	0.942	0.552	0.296	0.173	0.413	0.014
	300	0.934	0.933	0.487	0.268	0.153	0.383	0.010
	350	0.925	0.925	0.446	0.247	0.137	0.359	0.007
	400	0.918	0.918	0.415	0.225	0.122	0.335	0.006
150	100	0.992	0.860	0.876	0.724	0.630	0.776	0.231
	150	1.000	0.944	0.909	0.658	0.556	0.713	0.157
	200	0.985	0.795	0.760	0.619	0.520	0.676	0.129
	250	0.981	0.796	0.721	0.586	0.485	0.643	0.100
	300	0.979	0.872	0.694	0.563	0.458	0.623	0.080
	350	0.977	0.936	0.666	0.542	0.437	0.602	0.065
	400	0.975	0.960	0.646	0.523	0.415	0.586	0.056
200	100	0.997	0.936	0.952	0.855	0.793	0.885	0.462
	150	0.998	0.914	0.914	0.803	0.738	0.835	0.361
	200	1.000	0.932	0.893	0.767	0.700	0.801	0.301
	250	0.993	0.876	0.847	0.735	0.668	0.774	0.278
	300	0.992	0.863	0.822	0.715	0.649	0.753	0.245
	350	0.991	0.851	0.799	0.697	0.631	0.736	0.214
	400	0.991	0.843	0.782	0.682	0.616	0.721	0.198
250	100	0.999	0.973	0.982	0.933	0.892	0.947	0.648
	150	0.999	0.963	0.965	0.898	0.850	0.917	0.559
	200	0.999	0.953	0.948	0.870	0.820	0.892	0.497
	250	1.000	0.951	0.933	0.847	0.794	0.871	0.448
	300	0.997	0.933	0.911	0.827	0.770	0.856	0.431
	350	0.996	0.925	0.897	0.813	0.755	0.842	0.398
	400	0.996	0.919	0.884	0.798	0.740	0.828	0.376
300	100	1.000	0.990	0.994	0.971	0.949	0.978	0.782
	150	1.000	0.985	0.987	0.954	0.927	0.963	0.708
	200	1.000	0.981	0.980	0.936	0.903	0.948	0.654
	250	1.000	0.976	0.972	0.922	0.887	0.936	0.611
	300	1.000	0.973	0.963	0.908	0.870	0.924	0.574
	350	0.998	0.967	0.954	0.900	0.855	0.918	0.559
	400	0.998	0.963	0.946	0.889	0.842	0.908	0.537
350	100	1.000	0.996	0.997	0.989	0.980	0.991	0.869
	150	1.000	0.995	0.995	0.981	0.969	0.985	0.812
	200	1.000	0.992	0.992	0.971	0.955	0.977	0.767
	250	1.000	0.990	0.989	0.964	0.944	0.971	0.733
	300	1.000	0.988	0.985	0.957	0.936	0.966	0.705
	350	1.000	0.987	0.982	0.949	0.925	0.959	0.677
	400	0.999	0.984	0.976	0.946	0.916	0.956	0.659
400	100	1.000	0.999	0.999	0.995	0.991	0.996	0.930
	150	1.000	0.998	0.998	0.992	0.987	0.994	0.893
	200	1.000	0.997	0.997	0.989	0.981	0.991	0.852
	250	1.000	0.996	0.996	0.984	0.975	0.987	0.824
	300	1.000	0.996	0.995	0.981	0.971	0.985	0.801
	350	1.000	0.994	0.993	0.977	0.965	0.981	0.779
	400	1.000	0.994	0.992	0.974	0.960	0.979	0.757

<Table 4-6>: Average of the sensitivity with the SCAD under AR(2) situation

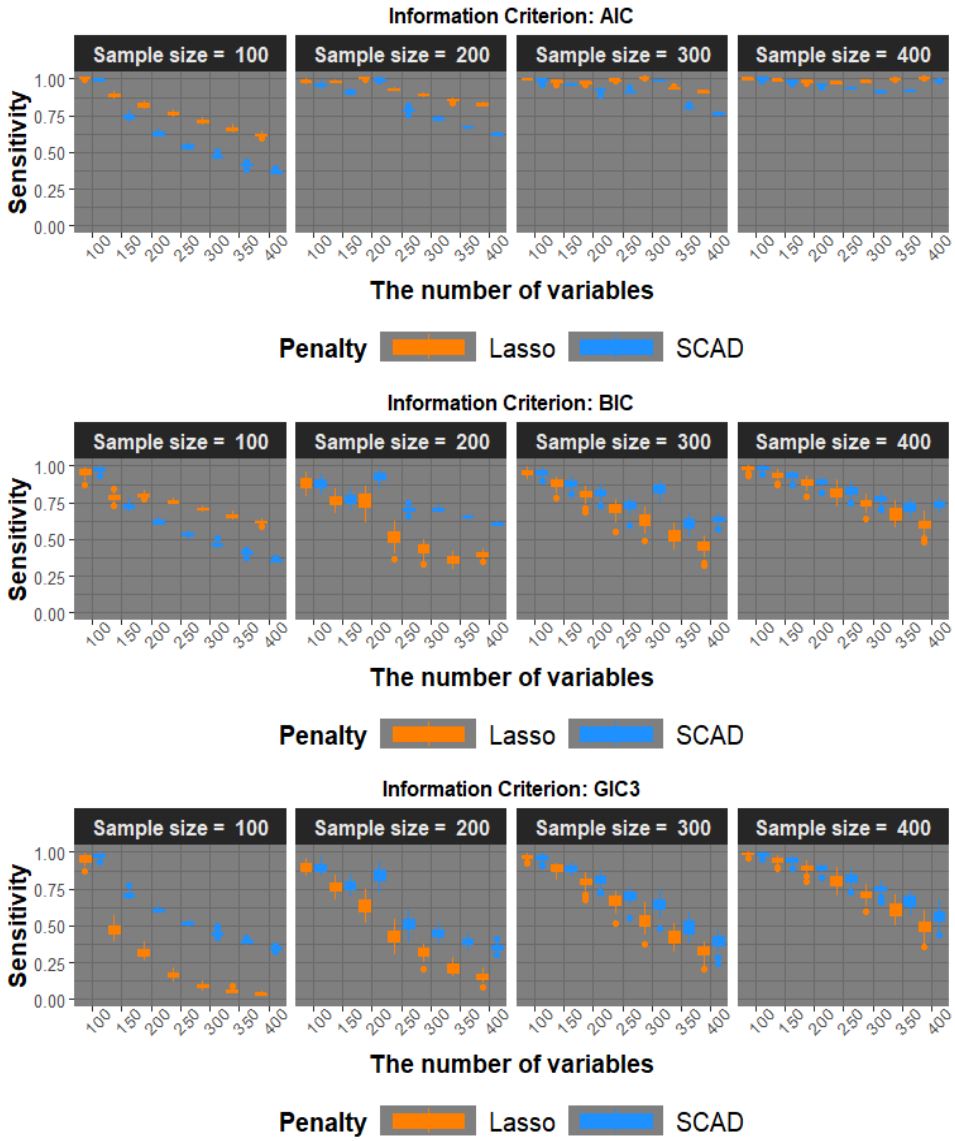
		Sensitivity						
n	p	AIC	BIC	GIC_3	GIC_4	GIC_5	GIC_6	GIC_7
100	100	0.996	0.991	0.991	0.744	0.597	0.896	0.210
	150	0.935	0.931	0.916	0.654	0.511	0.748	0.148
	200	0.917	0.916	0.906	0.591	0.439	0.725	0.110
	250	0.901	0.899	0.891	0.537	0.386	0.720	0.089
	300	0.886	0.885	0.876	0.494	0.345	0.714	0.073
	350	0.874	0.872	0.861	0.458	0.314	0.702	0.063
	400	0.861	0.860	0.845	0.424	0.285	0.686	0.056
150	100	0.985	0.928	0.935	0.883	0.851	0.900	0.550
	150	0.996	0.988	0.983	0.861	0.819	0.890	0.426
	200	0.971	0.947	0.904	0.841	0.794	0.864	0.363
	250	0.968	0.963	0.902	0.826	0.770	0.851	0.313
	300	0.964	0.962	0.903	0.814	0.751	0.843	0.270
	350	0.959	0.958	0.897	0.800	0.730	0.832	0.241
	400	0.954	0.953	0.890	0.786	0.711	0.822	0.220
200	100	0.996	0.967	0.973	0.937	0.922	0.947	0.805
	150	0.991	0.961	0.961	0.923	0.906	0.934	0.728
	200	0.997	0.989	0.978	0.915	0.898	0.926	0.660
	250	0.987	0.952	0.938	0.903	0.883	0.914	0.614
	300	0.986	0.964	0.931	0.897	0.877	0.908	0.564
	350	0.986	0.976	0.924	0.892	0.871	0.903	0.523
	400	0.984	0.980	0.917	0.887	0.865	0.898	0.489
250	100	0.999	0.987	0.990	0.971	0.958	0.977	0.902
	150	0.998	0.984	0.985	0.961	0.946	0.969	0.873
	200	0.993	0.980	0.979	0.953	0.940	0.961	0.847
	250	0.997	0.992	0.982	0.948	0.935	0.956	0.823
	300	0.994	0.976	0.968	0.939	0.923	0.948	0.792
	350	0.994	0.976	0.964	0.936	0.921	0.945	0.767
	400	0.994	0.978	0.959	0.931	0.916	0.940	0.744
300	100	1.000	0.995	0.996	0.988	0.980	0.991	0.941
	150	0.999	0.994	0.995	0.983	0.973	0.986	0.927
	200	0.999	0.993	0.993	0.979	0.968	0.984	0.915
	250	0.995	0.992	0.991	0.975	0.964	0.980	0.906
	300	0.997	0.996	0.990	0.971	0.960	0.977	0.894
	350	0.998	0.991	0.986	0.967	0.954	0.974	0.880
	400	0.998	0.990	0.984	0.963	0.950	0.970	0.870
350	100	1.000	0.998	0.998	0.995	0.992	0.996	0.966
	150	1.000	0.998	0.998	0.994	0.990	0.995	0.957
	200	1.000	0.997	0.997	0.992	0.985	0.994	0.947
	250	0.999	0.997	0.997	0.990	0.982	0.992	0.941
	300	0.996	0.997	0.996	0.988	0.980	0.991	0.936
	350	0.997	0.997	0.995	0.986	0.977	0.989	0.931
	400	0.999	0.996	0.994	0.985	0.974	0.988	0.921
400	100	1.000	0.999	0.999	0.998	0.997	0.999	0.981
	150	1.000	0.999	0.999	0.998	0.996	0.998	0.974
	200	1.000	0.999	0.999	0.997	0.994	0.998	0.968
	250	1.000	0.999	0.999	0.996	0.992	0.997	0.963
	300	0.999	0.999	0.999	0.996	0.992	0.997	0.961
	350	0.997	0.999	0.998	0.995	0.990	0.996	0.956
	400	0.998	0.999	0.998	0.994	0.989	0.995	0.953

<Table 4-7>: Average of the specificity with the LASSO under AR(2) situation

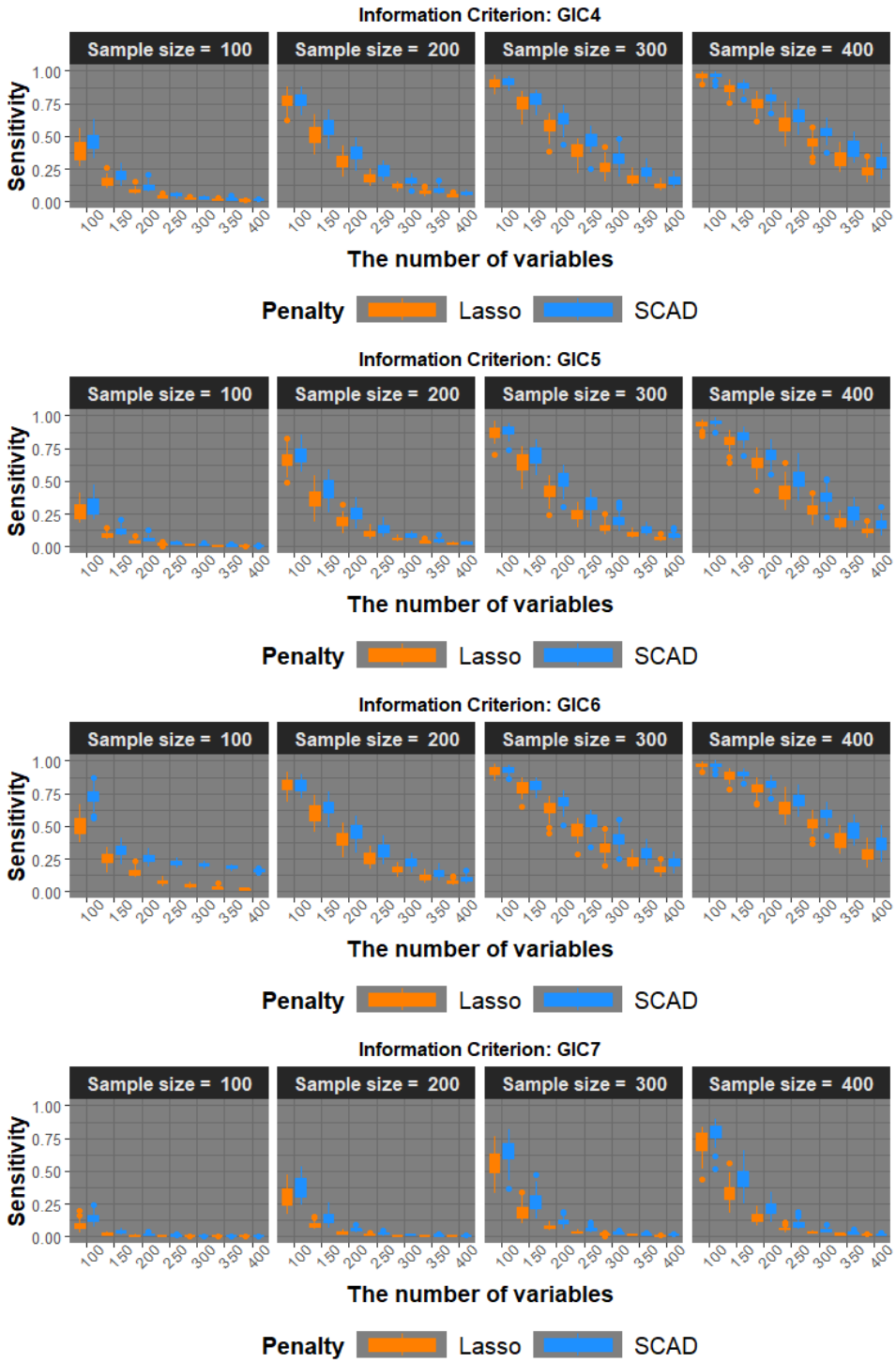
		Specificity						
n	p	AIC	BIC	GIC_3	GIC_4	GIC_5	GIC_6	GIC_7
100	100	0.000	0.083	0.087	0.994	0.998	0.972	1.000
	150	0.222	0.487	0.920	0.998	0.999	0.994	1.000
	200	0.347	0.366	0.925	0.999	1.000	0.997	1.000
	250	0.442	0.446	0.968	0.999	1.000	0.998	1.000
	300	0.515	0.517	0.992	1.000	1.000	0.999	1.000
	350	0.571	0.573	0.997	1.000	1.000	0.999	1.000
	400	0.616	0.618	0.998	1.000	1.000	0.999	1.000
150	100	0.325	0.955	0.946	0.987	0.994	0.980	1.000
	150	0.000	0.302	0.465	0.994	0.997	0.990	1.000
	200	0.184	0.978	0.986	0.996	0.998	0.994	1.000
	250	0.272	0.912	0.991	0.998	0.999	0.996	1.000
	300	0.350	0.711	0.993	0.998	0.999	0.997	1.000
	350	0.414	0.576	0.995	0.999	0.999	0.998	1.000
	400	0.468	0.538	0.996	0.999	1.000	0.998	1.000
200	100	0.570	0.952	0.938	0.982	0.990	0.976	0.999
	150	0.174	0.968	0.968	0.991	0.995	0.987	1.000
	200	0.000	0.643	0.892	0.994	0.997	0.992	1.000
	250	0.171	0.983	0.988	0.996	0.998	0.994	1.000
	300	0.236	0.986	0.991	0.997	0.998	0.996	1.000
	350	0.298	0.988	0.993	0.998	0.999	0.997	1.000
	400	0.353	0.982	0.995	0.998	0.999	0.997	1.000
250	100	0.635	0.951	0.935	0.978	0.986	0.973	0.998
	150	0.541	0.967	0.965	0.988	0.992	0.985	0.999
	200	0.072	0.976	0.978	0.992	0.995	0.990	1.000
	250	0.000	0.870	0.981	0.995	0.997	0.993	1.000
	300	0.167	0.985	0.989	0.996	0.998	0.995	1.000
	350	0.217	0.988	0.992	0.997	0.998	0.996	1.000
	400	0.268	0.989	0.993	0.997	0.998	0.996	1.000
300	100	0.662	0.951	0.933	0.976	0.983	0.971	0.996
	150	0.655	0.967	0.963	0.987	0.990	0.983	0.998
	200	0.469	0.975	0.976	0.991	0.994	0.989	0.999
	250	0.028	0.980	0.983	0.993	0.995	0.992	0.999
	300	0.000	0.964	0.987	0.995	0.996	0.993	1.000
	350	0.167	0.987	0.991	0.996	0.997	0.995	1.000
	400	0.207	0.989	0.992	0.997	0.998	0.996	1.000
350	100	0.677	0.952	0.933	0.974	0.981	0.970	0.995
	150	0.698	0.967	0.962	0.985	0.989	0.982	0.997
	200	0.658	0.975	0.975	0.990	0.993	0.988	0.999
	250	0.360	0.980	0.982	0.993	0.995	0.991	0.999
	300	0.012	0.984	0.986	0.994	0.996	0.993	0.999
	350	0.000	0.984	0.989	0.995	0.997	0.994	1.000
	400	0.168	0.989	0.992	0.996	0.997	0.995	1.000
400	100	0.686	0.953	0.933	0.974	0.981	0.970	0.993
	150	0.717	0.968	0.962	0.985	0.988	0.982	0.996
	200	0.717	0.976	0.975	0.989	0.992	0.988	0.998
	250	0.633	0.981	0.982	0.992	0.994	0.991	0.999
	300	0.235	0.984	0.986	0.994	0.995	0.992	0.999
	350	0.006	0.986	0.989	0.995	0.996	0.994	0.999
	400	0.000	0.988	0.991	0.996	0.997	0.995	0.999

<Table 4-8>: Average of the specificity with the SCAD under AR(2) situation

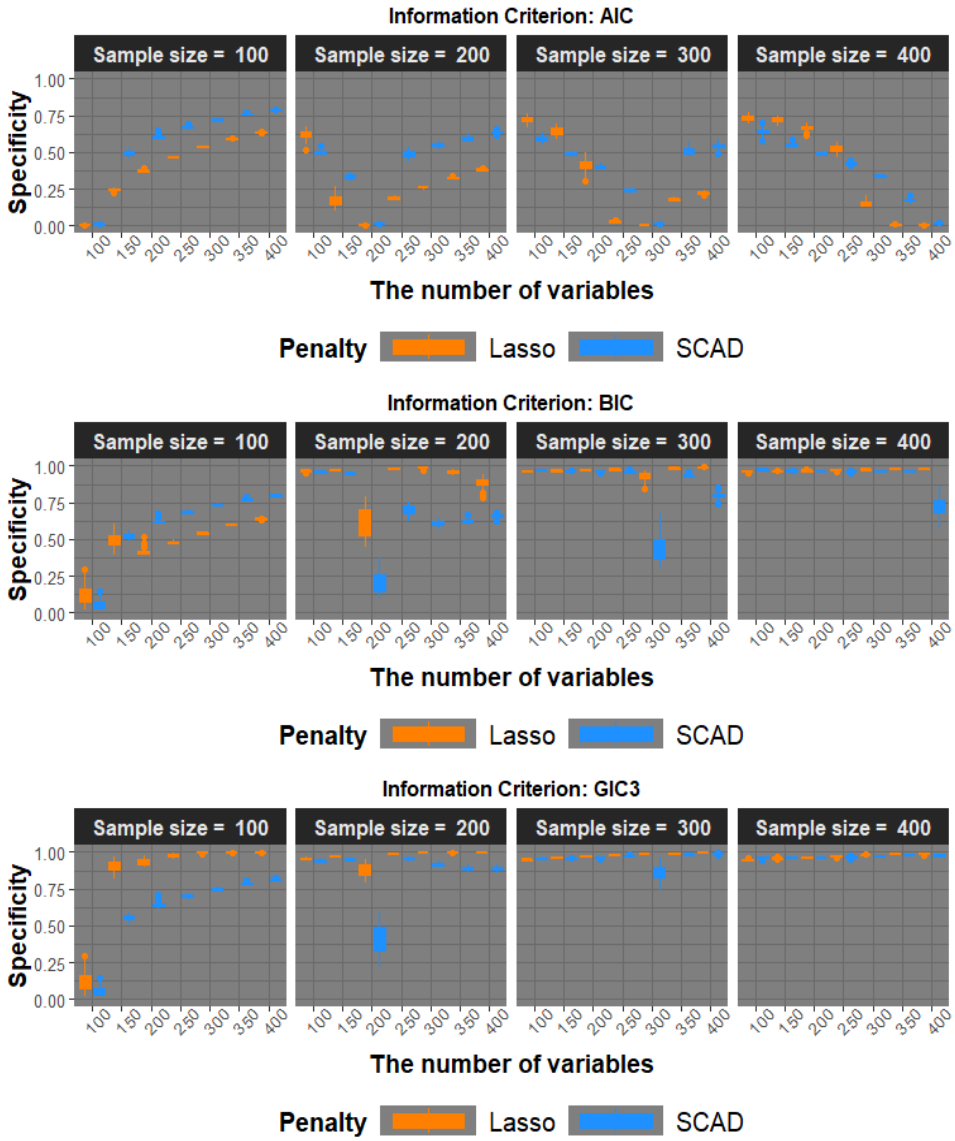
		Specificity						
n	p	AIC	BIC	GIC_3	GIC_4	GIC_5	GIC_6	GIC_7
100	100	0.007	0.045	0.046	0.925	0.996	0.498	1.000
	150	0.461	0.491	0.559	0.994	0.998	0.981	1.000
	200	0.577	0.593	0.627	0.996	0.999	0.962	1.000
	250	0.654	0.669	0.696	0.997	0.999	0.947	1.000
	300	0.708	0.722	0.749	0.998	0.999	0.942	1.000
	350	0.748	0.761	0.789	0.998	1.000	0.944	1.000
	400	0.777	0.789	0.820	0.999	1.000	0.948	1.000
150	100	0.375	0.965	0.955	0.989	0.994	0.984	1.000
	150	0.007	0.102	0.159	0.993	0.996	0.909	1.000
	200	0.456	0.687	0.933	0.995	0.998	0.993	1.000
	250	0.537	0.592	0.887	0.996	0.998	0.995	1.000
	300	0.602	0.627	0.875	0.997	0.999	0.996	1.000
	350	0.652	0.670	0.887	0.998	0.999	0.996	1.000
	400	0.692	0.707	0.902	0.998	0.999	0.997	1.000
200	100	0.476	0.973	0.964	0.990	0.994	0.986	0.999
	150	0.308	0.981	0.981	0.994	0.997	0.992	1.000
	200	0.007	0.201	0.403	0.996	0.998	0.993	1.000
	250	0.471	0.951	0.991	0.997	0.998	0.996	1.000
	300	0.529	0.808	0.993	0.998	0.999	0.997	1.000
	350	0.579	0.707	0.993	0.998	0.999	0.997	1.000
	400	0.622	0.685	0.993	0.998	0.999	0.998	1.000
250	100	0.539	0.977	0.969	0.990	0.994	0.987	0.999
	150	0.407	0.983	0.982	0.994	0.997	0.992	1.000
	200	0.258	0.987	0.988	0.996	0.998	0.994	1.000
	250	0.008	0.329	0.674	0.997	0.998	0.996	1.000
	300	0.491	0.990	0.994	0.998	0.999	0.997	1.000
	350	0.533	0.974	0.995	0.998	0.999	0.997	1.000
	400	0.573	0.916	0.996	0.998	0.999	0.998	1.000
300	100	0.587	0.981	0.974	0.990	0.994	0.988	0.999
	150	0.461	0.985	0.984	0.994	0.996	0.992	0.999
	200	0.365	0.988	0.989	0.996	0.997	0.994	1.000
	250	0.220	0.990	0.992	0.997	0.998	0.996	1.000
	300	0.008	0.490	0.891	0.997	0.998	0.997	1.000
	350	0.511	0.993	0.995	0.998	0.999	0.997	1.000
	400	0.542	0.993	0.996	0.998	0.999	0.998	1.000
350	100	0.624	0.985	0.979	0.991	0.994	0.989	0.999
	150	0.509	0.988	0.986	0.994	0.996	0.993	0.999
	200	0.416	0.990	0.990	0.995	0.997	0.995	1.000
	250	0.333	0.991	0.992	0.996	0.998	0.996	1.000
	300	0.188	0.992	0.994	0.997	0.998	0.996	1.000
	350	0.008	0.636	0.974	0.998	0.999	0.997	1.000
	400	0.529	0.994	0.996	0.998	0.999	0.998	1.000
400	100	0.654	0.988	0.983	0.992	0.995	0.991	0.999
	150	0.548	0.990	0.989	0.995	0.997	0.994	1.000
	200	0.452	0.992	0.991	0.996	0.997	0.995	1.000
	250	0.386	0.993	0.993	0.997	0.998	0.996	1.000
	300	0.305	0.994	0.994	0.997	0.998	0.997	1.000
	350	0.162	0.994	0.995	0.998	0.999	0.997	1.000
	400	0.009	0.774	0.994	0.998	0.999	0.998	1.000



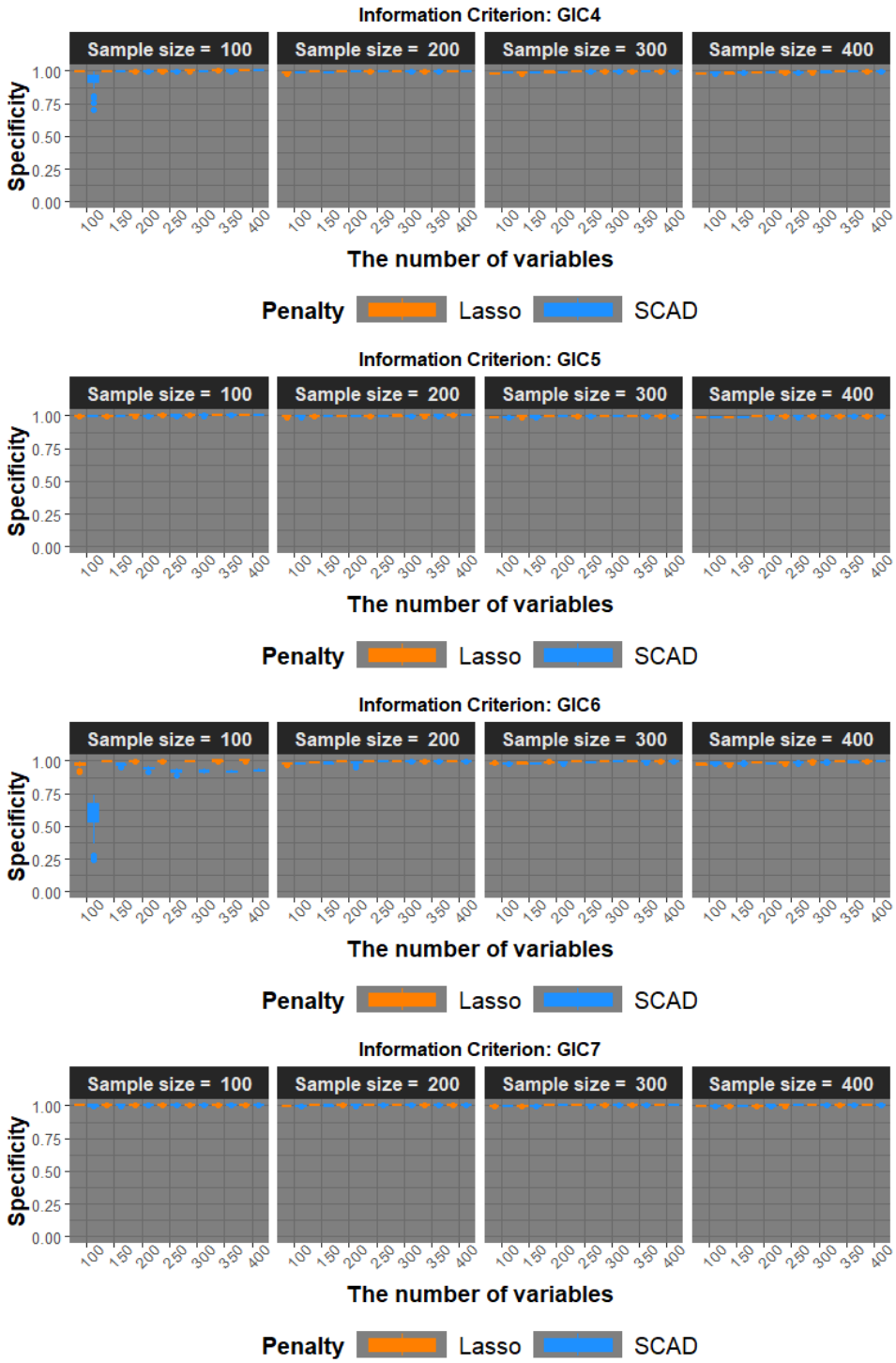
<Figure 4-1> Sensitivity of AIC, BIC, and GIC_3 when sample size = 100,200,300, and 400 in the Model 1



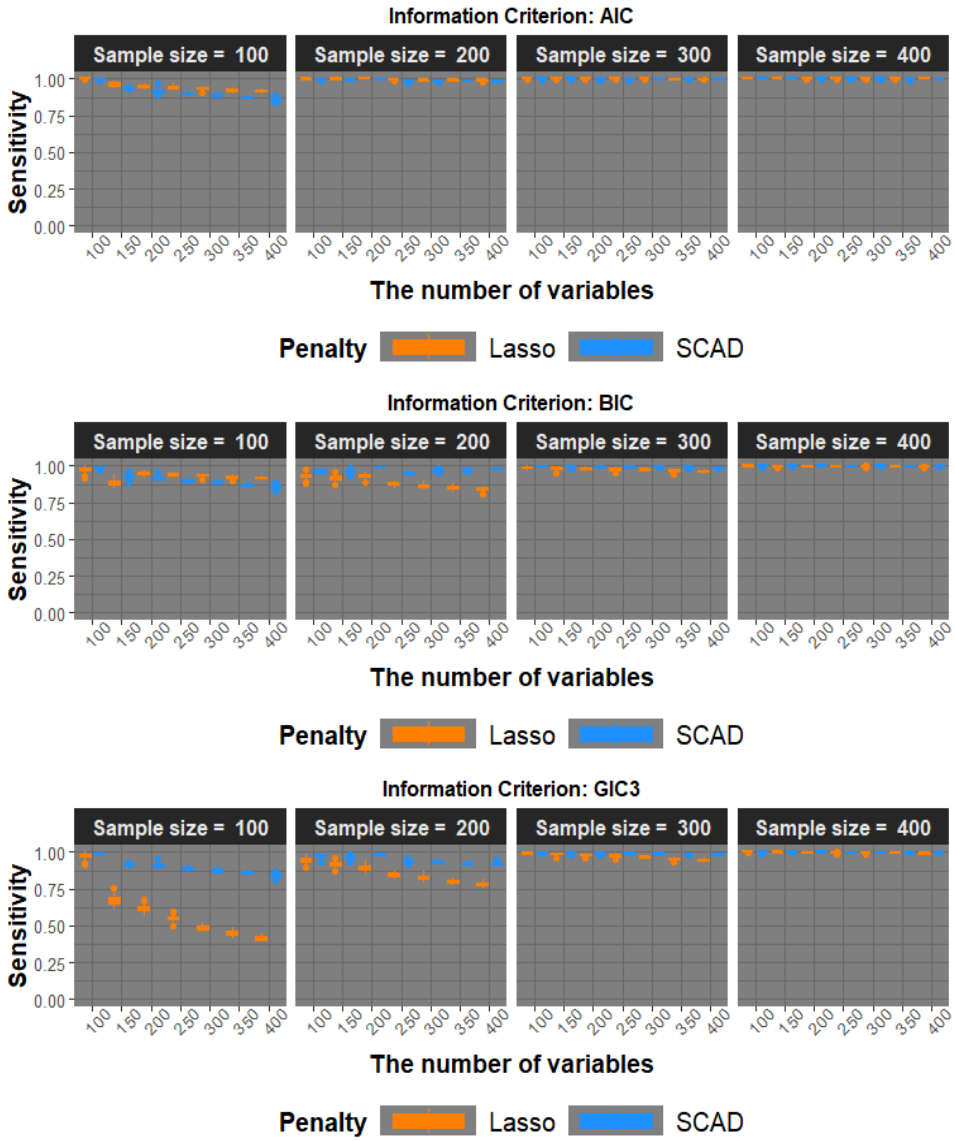
<Figure 4-2> Sensitivity of GIC_4, GIC_5, GIC_6, and GIC_7 when sample size = 100,200,300, and 400 in the Model 1



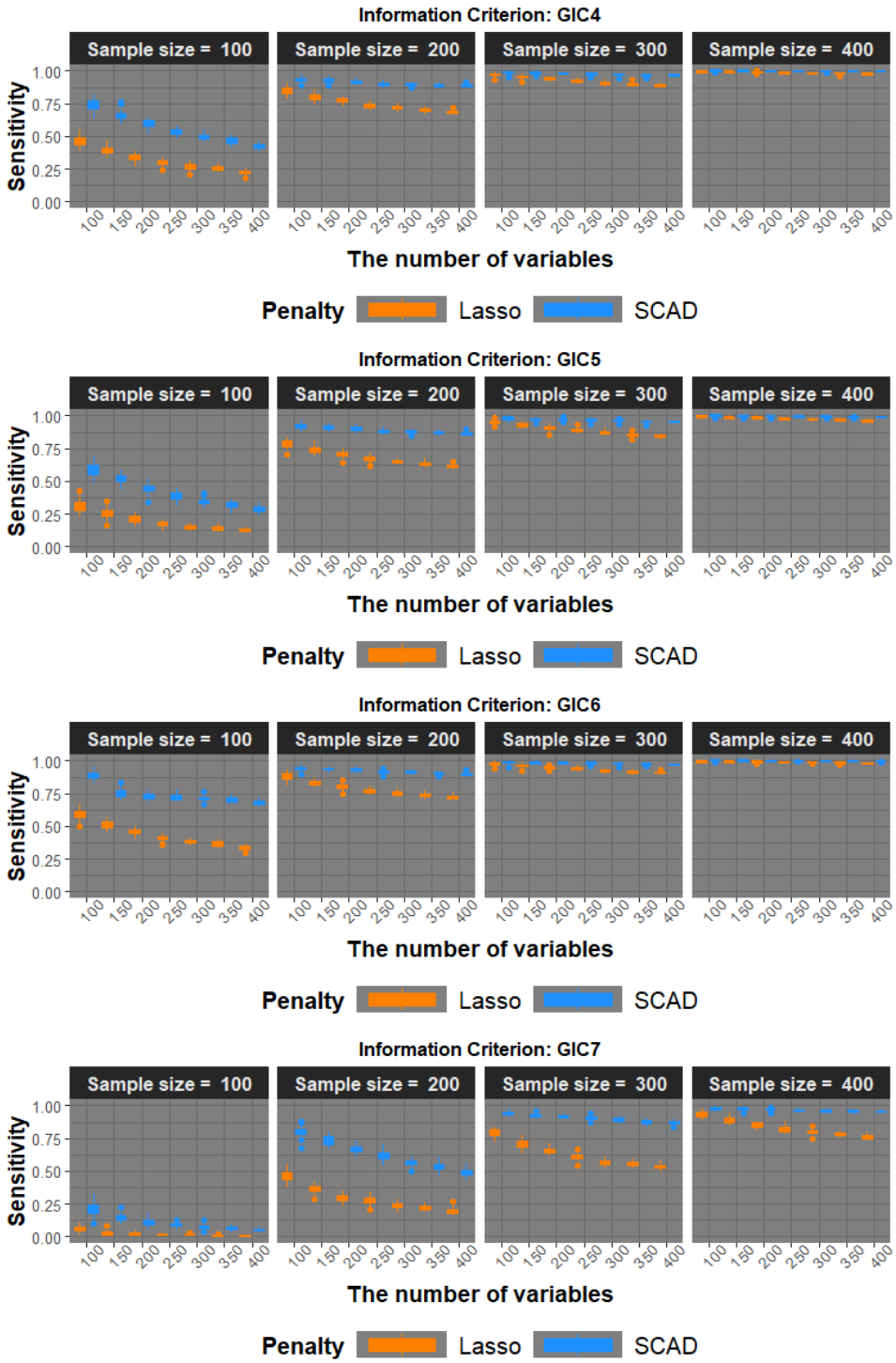
<Figure 4-3> Specificity of AIC, BIC, and GIC_3 when sample size = 100,200,300, and 400 in the Model 1



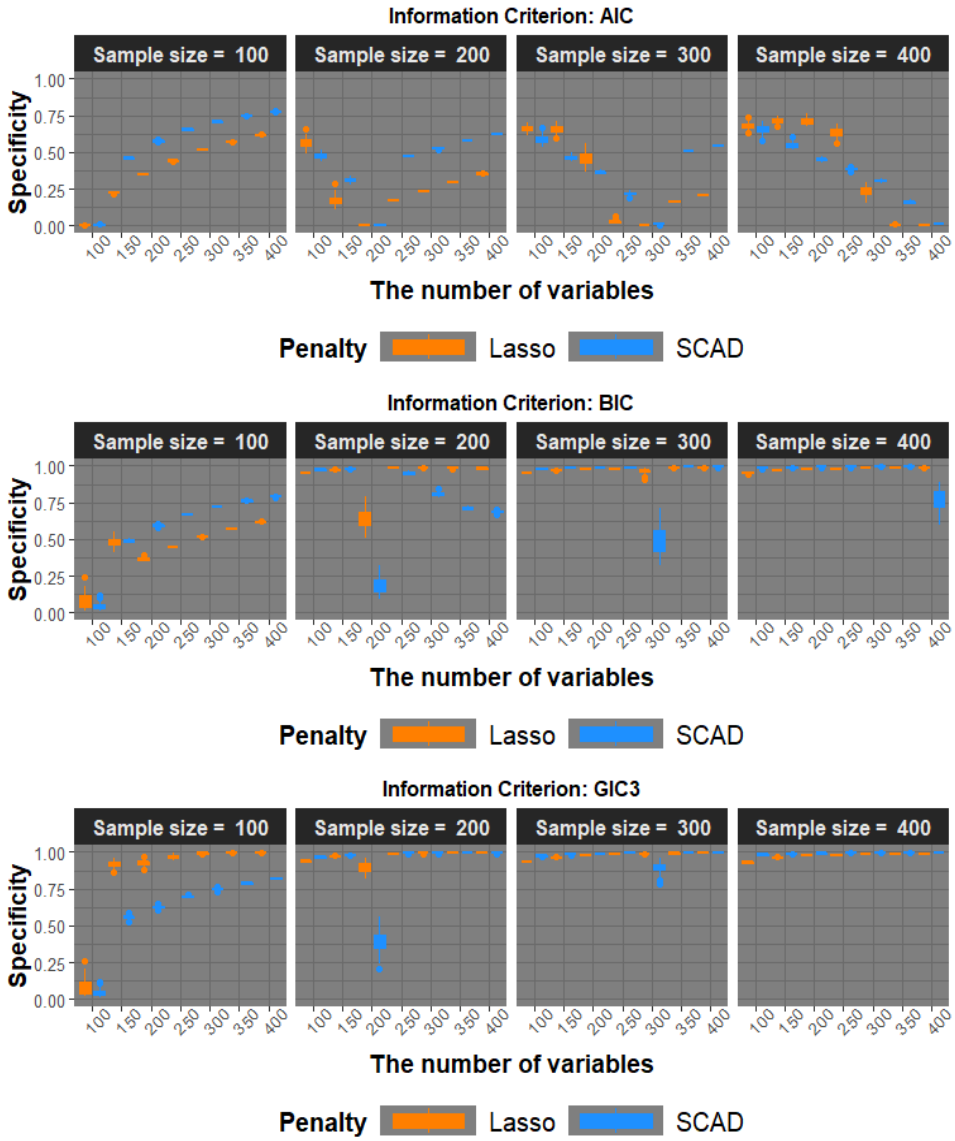
<Figure 4-4> Specificity of GIC_4, GIC_5, GIC_6, and GIC_7 when sample size = 100,200,300, and 400 in the Model 1



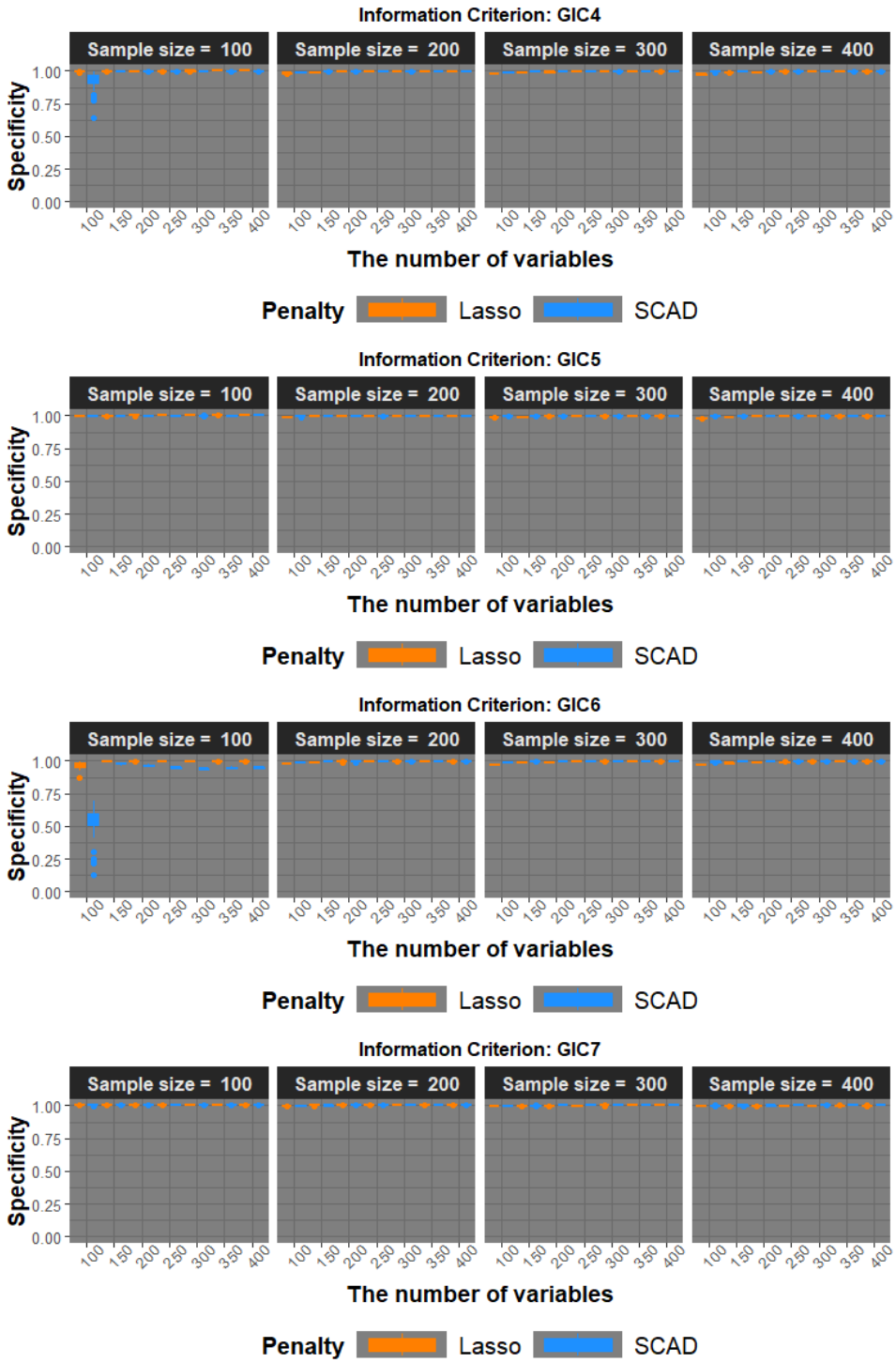
<Figure 4-5> Sensitivity of AIC, BIC, and GIC_3 when sample size = 100,200,300, and 400 in the AR(2) model



<Figure 4-6> Sensitivity of GIC_4, GIC_5, GIC_6, and GIC_7 when sample size = 100,200,300, and 400 in the AR(2) model



<Figure 4-7> Specificity of AIC, BIC, and GIC_3 when sample size = 100,200,300, and 400 in the AR(2) model



<Figure 4-8> Specificity of GIC_4, GIC_5, GIC_6, and GIC_7 when sample size = 100,200,300, and 400 in the AR(2) model

Chapter 5. Conclusion

In this paper, we studied diverse model selection criteria based on the GIC for the neighborhood selection by the simulation study. Depending on the n and p, the GICs indicate different results and patterns in terms of sensitivity and specificity. The advantage of comparison of various model selection criteria lies in its diversity of choices about how to select the tuning parameters under the specific n and p. However, there is a lack of theoretical properties about the selection consistency for the GGMs. And this study is limited to the neighborhood selection method. We believe that comparison of GICs for another penalized estimator, for example GLASSO, is valuable.

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Abstract(in Korean)

가우스 그래프 모형에서 이웃 선택 방법의 모형 선택 기준에 대한 비교

고건
응용통계학과
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일반적으로 고차원 상황에서 가우스 그래프 모형을 추정하기 위해서 벌점 추정기가 사용된다. 대표적인 추정기가 이웃 선택 방법이다. 이웃 선택 방식은 벌점 회귀 기반의 방식이기 때문에 매개변수 조정에 대한 모형 선택 문제가 있다. 본 논문에서는 가우스 그래프 모형에서 이웃 선택 방식에 대해 일반화 정보 기준에 기반한 다양한 모형 선택 기준을 시뮬레이션 연구를 통해 비교한다.

주제어 : 가우스 그래피컬 모형, 모형 선택 기준, 이웃 선택 방법.